

Customer Shopping Behaviour Analysis

1. Project Background and Objective

This project focuses on analysing customer shopping behaviour using transactional data to understand how customers make purchasing decisions. The main objective is to identify patterns in customer spending, product preferences, and buying habits that can help the company improve its business strategies.

By analysing customer data, the project aims to discover important factors affecting purchases, such as customer demographics, product categories, discounts, reviews, and subscription behaviour. These insights can support better decision-making and help improve customer satisfaction and business growth.

2. Understanding the Dataset

The dataset contains customer transaction information that helps analyse different aspects of shopping behaviour. It includes details related to customer demographics, purchased products, spending patterns, and shopping preferences.

- Rows: 3,900
- Columns: 18

The main information available in the dataset includes:

- Customer details such as age, gender, location, and subscription status.
- Purchase information including product category, item purchased, purchase amount, season, size, and colour.
- Shopping behaviour details such as discount usage, promotional offers, previous purchases, purchase frequency, customer reviews, and shipping preferences.

This information provides a complete view of customer interactions and helps identify trends within different customer groups.

3. Data Preparation and Cleaning Process

Before performing analysis, the raw dataset was prepared and cleaned using Python. Data preparation is an important step because accurate analysis depends on having reliable and consistent data.

The cleaning process included checking the dataset structure, identifying missing values, improving column names, creating useful new features, and ensuring data consistency.

Missing values in customer review ratings were handled by applying an appropriate method to maintain data quality. Additional features were also created to better understand customers, such as age groups and purchasing frequency patterns.

[4]: `# Summary statistics using .describe()`
`df.describe(include='all')`

[4]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

4. Customer Behaviour Analysis Using SQL

After cleaning the data, SQL was used to analyse customer transactions and answer important business questions.

The analysis focused on understanding:

- Revenue contribution from different customer groups.
- Spending behaviour of customers using discounts.
- Product performance based on customer ratings.
- Differences between shipping methods.
- Purchasing patterns of subscribers and non-subscribers.
- Customer loyalty and repeat purchasing behaviour.

These analyses help identify customer trends and provide insights that can support business planning.

--What is the total revenue generated by male & female customers?

	gender text	revenue numeric
1	Female	75191
2	Male	157890

--Which customers used a discount but still spent more than the average purchase amount?

Data Output

Messages

Notifications

SQL

	customer_id bigint		purchase_amount bigint	
1	2		64	
2	3		73	
3	4		90	
4	7		85	
5	9		97	
6	12		68	
7	13		72	
8	16		81	

-- Which are the top 5 products with the highest average review rating?

Data Output Messages Notifications		
		
		
		
		
	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

--Compare the average Purchase Amounts between Standard and Express Shipping.

Data Output Messages Notifications		
		
		
		
		
	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

--Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

--Which 5 products have the highest percentage of purchases with discounts applied?

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

--Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

--What are the top 3 most purchased products within each category?

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

--Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

--What is the revenue contribution of each age group?

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Customer Segmentation and Purchase Insights

Customer segmentation was performed to classify customers according to their purchasing behaviour. This helps the company understand different customer groups and create more targeted strategies.

Customers were analysed based on their purchase history and divided into different categories, such as:

- New customers
- Returning customers
- Loyal customers

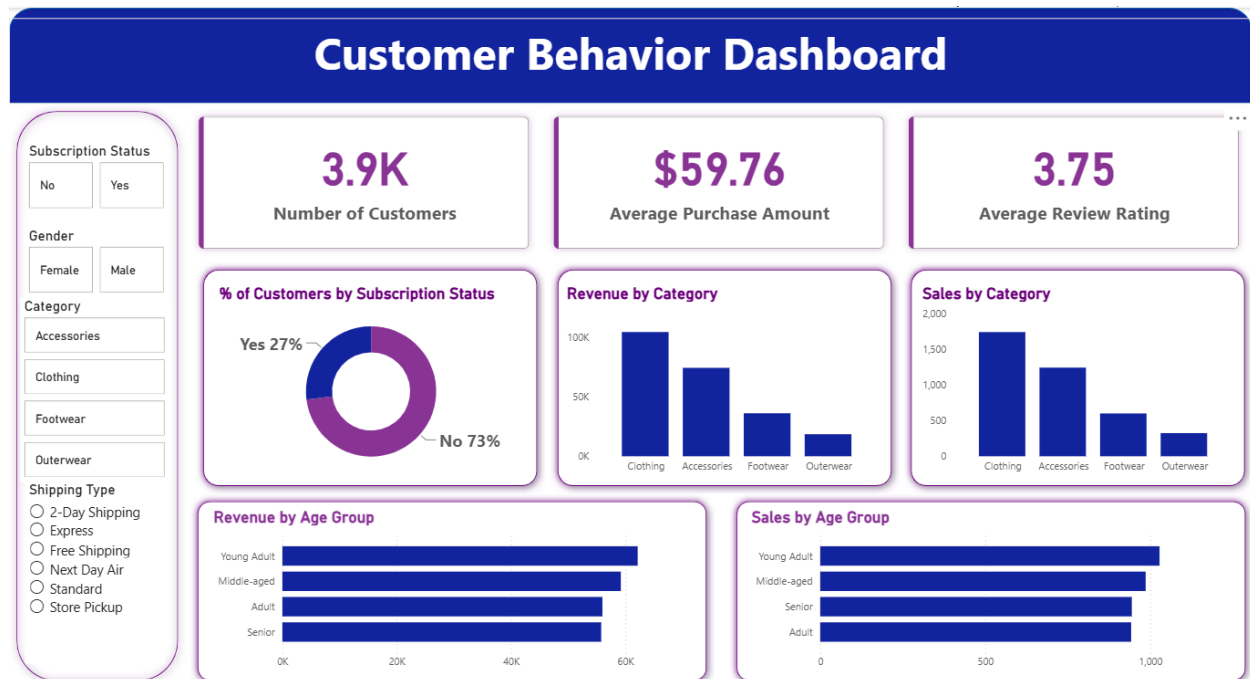
The analysis also explored product popularity, repeat buyers, subscription patterns, and revenue contribution across different age groups.

6. Interactive Dashboard Development Using Power BI

The final stage involved creating an interactive Power BI dashboard to present important findings visually.

The dashboard helps stakeholders easily understand customer behaviour through visual representations of key information, including customer numbers, average purchase amount, ratings, revenue trends, product categories, and customer segments.

By presenting data in an interactive format, the dashboard allows users to explore patterns and make informed business decisions.



7. Business Recommendations

Based on the analysis results, several recommendations can be suggested to improve business performance.

The company can focus on:

- Encouraging customer subscriptions by offering valuable benefits.
- Developing loyalty programs to retain repeat customers.
- Reviewing discount strategies to increase sales while maintaining profitability.
- Promoting highly rated and popular products.
- Using targeted marketing campaigns for specific customer groups.

These recommendations can help the company improve customer engagement, increase sales opportunities, and build stronger relationships with customers.